

41. Let S be the set of all values of $\theta \in [-\pi, \pi]$ for which the system of linear equations

$$x + y + \sqrt{3}z = 0$$

$$-x + (\tan \theta)y + \sqrt{7}z = 0$$

$$x + y + (\tan \theta)z = 0$$

has non-trivial solution. Then $\frac{120}{\pi} \sum_{\theta \in S} \theta$ is equal to :

माना $\theta \in [-\pi, \pi]$ के सभी मानों, जिनके लिए रैखिक समीकरण निकाय

$$x + y + \sqrt{3}z = 0$$

$$-x + (\tan \theta)y + \sqrt{7}z = 0$$

$$x + y + (\tan \theta)z = 0$$

का अतुच्छ हल है, का समुच्च S है। तो $\frac{120}{\pi} \sum_{\theta \in S} \theta$ बराबर है :

(A) 30

(B) 20

(C) 10

(D) 40

Ans. B

Sol. For non trivial solutions

$$D = 0$$

$$\begin{vmatrix} 1 & 1 & \sqrt{3} \\ -1 & \tan \theta & \sqrt{7} \\ 1 & 1 & \tan \theta \end{vmatrix} = 0$$

$$\tan^2 \theta - (\sqrt{3} - 1)\tan \theta - \sqrt{3} = 0$$

$$\tan \theta = \sqrt{3}, -1$$

$$\theta = \left\{ \frac{\pi}{3}, \frac{-2\pi}{3}, \frac{-\pi}{4}, \frac{3\pi}{4} \right\}$$

$$\frac{120}{\pi} (\sum \theta) = \frac{120}{\pi} \times \frac{\pi}{6} = 20$$

42. Let A be a 3×3 real matrix such that $A \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, $A \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = 4 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$, $A \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$. Then, the system

$$(A - 3I) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \text{ has :}$$

(A) no solution

(B) exactly two solutions

(C) infinitely many solutions

(D) unique solution

माना एक 3×3 वास्तविक आव्यूह A के लिए $A \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, A \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = 4 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, A \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ हैं। तो निकाय

$$(A - 3I) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} :$$

(A) कोई हल नहीं है

(B) के मात्र दो हल हैं

(C) के अनंत हल हैं

(D) का अद्वितीय हल है

Ans. D

Sol. $A \rightarrow 3 \times 3$ matrix

$$A \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$A \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = 4 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$A \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\text{Then } (A - 3I) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\text{Let } A = \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}$$

Now using given condition

$$(i) \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}_{3 \times 3} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}_{3 \times 1} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} \Rightarrow \begin{matrix} y_1 = 0 \\ y_2 = 2 \\ y_3 = 0 \end{matrix}$$

$$(ii) \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}_{3 \times 3} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 \\ 0 \\ 4 \end{pmatrix}$$

$$-x_1 + z_1 = -4$$

$$-x_2 + z_2 = 0$$

$$-x_3 + z_3 = 4$$

$$(iii) \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}_{3 \times 3} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$$

$$x_1 + z_1 = 2$$

$$x_2 + z_2 = 0$$

$$x_3 + z_3 = 2$$

$$-x_1 + z_1 = -4$$

$$x_1 + z_1 = 2$$

$$\Rightarrow z_1 = -1$$

$$x_1 = 3$$

So now

$$A = \begin{bmatrix} 3 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 3 \end{bmatrix}$$

$$A - 3I = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$(A - 3I) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$-z = 3 \Rightarrow z = -3$$

$$-y = 2 \Rightarrow y = -2 \quad \Rightarrow \text{unique Solution}$$

$$-x = 1 \Rightarrow x = -1$$

43. If A is a square matrix of order 3 such that $\det(A) = 3$ and $\det\left(\text{adj}\left(-4 \text{adj}\left(-3 \text{adj}\left(3 \text{adj}\left((2A)^{-1}\right)\right)\right)\right)\right) = 2^m 3^n$, then $m + 2n$ is equal to :

यदि A कोटि 3 का एक वर्ग आव्यूह इस प्रकार है कि $\det(A) = 3$ है तथा

$\det\left(\text{adj}\left(-4 \text{adj}\left(-3 \text{adj}\left(3 \text{adj}\left((2A)^{-1}\right)\right)\right)\right)\right) = 2^m 3^n$, तो $m + 2n$ बराबर है :

- (A) 3 (B) 2 (C) 6 (D) 4

Ans. D

Sol.

$$\begin{aligned} & \left| \text{adj}\left(-4 \text{adj}\left(-3 \text{adj}\left(3 \text{adj}\left(2A\right)^{-1}\right)\right)\right) \right| \\ &= \left| -4 \text{adj}\left(-3 \text{adj}\left(3 \text{adj}\left(2A\right)^{-1}\right)\right) \right|^2 \\ &= (-4)^6 \left| \text{adj}\left(-3 \text{adj}\left(3 \text{adj}\left(2A\right)^{-1}\right)\right) \right|^2 \end{aligned}$$

$$\begin{aligned}
 &= 4^6 \cdot (-3)^{12} \left| \text{adj}(3\text{adj}(2A)^{-1}) \right|^4 \\
 &= 4^6 \cdot 3^{12} \cdot 3^{24} \left| \text{adj}(2A)^{-1} \right|^8 \\
 &= 4^6 \cdot 3^{36} \left| (2A)^{-1} \right|^{16} \\
 &= 4^6 \cdot 3^{36} \cdot 2^{-48} \left(\frac{1}{|A|} \right)^{16} = \frac{4^6 \cdot 3^{36} \cdot 2^{-48}}{3^{16}} = 2^{-36} \cdot 3^{20}
 \end{aligned}$$

$$\Rightarrow m = -36, n = 20$$

$$m + 2n = -36 + 40 = 4$$

44. Let $A = \begin{bmatrix} \alpha & -1 \\ 6 & \beta \end{bmatrix}$, $\alpha > 0$, such that $\det(A) = 0$ and $\alpha + \beta = 1$. If I denotes 2×2 identity matrix, then the matrix $(I + A)^8$ is :

माना $A = \begin{bmatrix} \alpha & -1 \\ 6 & \beta \end{bmatrix}$, $\alpha > 0$, $\det(A) = 0$ और $\alpha + \beta = 1$ हैं। यदि I , 2×2 का तत्समक आव्यूह है, तो आव्यूह $(I + A)^8$ है:

(A) $\begin{bmatrix} 257 & -64 \\ 514 & -127 \end{bmatrix}$ (B) $\begin{bmatrix} 1025 & -511 \\ 2024 & -1024 \end{bmatrix}$ (C) $\begin{bmatrix} 4 & -1 \\ 6 & -1 \end{bmatrix}$ (D) $\begin{bmatrix} 766 & -255 \\ 1530 & -509 \end{bmatrix}$

Ans. D

Sol. $|A| = 0$

$$\alpha\beta + 6 = 0$$

$$\alpha\beta = -6$$

$$\alpha + \beta = 1$$

$$\Rightarrow \alpha = 3, \beta = -2$$

$$A = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix}$$

$$\therefore A^2 = A$$

$$A = A^2 = A^3 = A^4 = A^5$$

$$(I + A)^8$$

$$= I + {}^8C_1 A^1 + {}^8C_2 A^2 + \dots + {}^8C_8 A^8$$

$$= I + A({}^8C_1 + {}^8C_2 + \dots + {}^8C_8)$$

$$\begin{aligned}
 &= I + A(2^8 - 1) \\
 &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 765 & -255 \\ 1530 & -510 \end{bmatrix} \\
 &= \begin{bmatrix} 766 & -255 \\ 1530 & -509 \end{bmatrix}
 \end{aligned}$$

45. Let $S = \left\{ x \in \mathbb{R} : (\sqrt{3} + \sqrt{2})^x + (\sqrt{3} - \sqrt{2})^x = 10 \right\}$. Then the number of elements in S is :

माना $S = \left\{ x \in \mathbb{R} : (\sqrt{3} + \sqrt{2})^x + (\sqrt{3} - \sqrt{2})^x = 10 \right\}$ है। तो S में अवयवों की संख्या है :

- (A) 4 (B) 0 (C) 1 (D) 2

Ans. D

Sol. $(\sqrt{3} + \sqrt{2})^x + (\sqrt{3} - \sqrt{2})^x = 10$

$$\text{Let } (\sqrt{3} + \sqrt{2})^x = t$$

$$t + \frac{1}{t} = 10$$

$$t^2 - 10t + 1 = 0$$

$$t = \frac{10 \pm \sqrt{100 - 4}}{2} = 5 \pm 2\sqrt{6}$$

$$(\sqrt{3} + \sqrt{2})^x = (\sqrt{3} \pm \sqrt{2})^2$$

$$x = 2 \text{ or } x = -2$$

Number of solutions = 2

46. If $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots = \frac{\pi^4}{90}$, $\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots = \alpha$, $\frac{1}{2^4} + \frac{1}{4^4} + \frac{1}{6^4} + \dots = \beta$, then $\frac{\alpha}{\beta}$ is equal to :

यदि $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots = \frac{\pi^4}{90}$, $\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots = \alpha$, $\frac{1}{2^4} + \frac{1}{4^4} + \frac{1}{6^4} + \dots = \beta$, है, तो $\frac{\alpha}{\beta}$ बराबर है :

- (A) 14 (B) 18 (C) 15 (D) 23

Ans. C

Sol. $\left(\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots\right) + \frac{1}{2^4} \left(\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots\right) = \frac{\pi^2}{90}$

$$\alpha + \frac{1}{2^4} \left(\frac{\pi^2}{90}\right) = \frac{\pi^2}{90}$$

$$\alpha = \frac{\pi^2}{90} \left(1 - \frac{1}{16}\right) = \frac{\pi^2 \cdot (15)}{90 \cdot 16} = \frac{\pi^2}{6 \times 16}$$

Now, $\beta = \frac{\pi^2}{90} \cdot \left(\frac{1}{2^4}\right)$

$$\frac{\alpha}{\beta} = 15$$

47. Let 2nd, 8th and 44th terms of a non-constant A. P. be respectively the 1st, 2nd and 3rd terms of a G.P. If the first term of the A.P. is 1, then the sum of its first 20 terms is equal to :

माना एक भिन्न पदों की A. P. का दूसरा, आठवाँ तथा चवालिसवाँ पद, एक G. P. के क्रमशः पहला, दूसरा तथा तीसरा पद हैं। यदि A. P. का प्रथम पद 1 है, तो इसके प्रथम 20 पदों का योग है :

- (A) 990 (B) 980 (C) 960 (D) 970

Ans. D

Sol. $a + d, a + 7d, a + 43d$ are 1st, 2nd, 3rd term of a G.P.

$$\frac{a + 7d}{a + d} = \frac{a + 43d}{a + 7d}$$

$$\Rightarrow (a + 7d)^2 = (a + 43d)(a + d)$$

$$\Rightarrow a^2 + 49d^2 + 14ad = a^2 + 44ad + 43d^2$$

$$\Rightarrow 6d^2 = 30ad$$

$$\Rightarrow d = 0, 5$$

But $d \neq 0$

$$\Rightarrow d = 5, a = 1$$

$$S_{20} = \frac{20}{2} [2 + (19)^5]$$

$$= 10(95 + 2) = 970$$

48. If the system of equations

$$(\lambda - 1)x + (\lambda - 4)y + \lambda z = 5$$

$$\lambda x + (\lambda - 1)y + (\lambda - 4)z = 7$$

$$(\lambda + 1)x + (\lambda + 2)y - (\lambda + 2)z = 9$$

has infinitely many solutions, then $\lambda^2 + \lambda$ is equal to :

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$$(\lambda - 1)x + (\lambda - 4)y + \lambda z = 5$$

$$\lambda x + (\lambda - 1)y + (\lambda - 4)z = 7$$

$$(\lambda + 1)x + (\lambda + 2)y - (\lambda + 2)z = 9$$

के अनन्त हल हैं, तो $\lambda^2 + \lambda$ बराबर है :

(A) 6

(B) 12

(C) 10

(D) 20

Ans. B

Sol. D = 0

$$\begin{vmatrix} \lambda - 1 & \lambda - 4 & \lambda \\ \lambda & \lambda - 1 & \lambda - 4 \\ \lambda + 1 & \lambda + 2 & -\lambda - 2 \end{vmatrix} = 0$$

$$R_1 \rightarrow R_1 - R_2 \text{ \& } R_2 \rightarrow R_2 - R_3$$

$$\begin{vmatrix} -1 & -3 & 4 \\ -1 & -3 & 2\lambda - 2 \\ \lambda + 1 & \lambda + 2 & -\lambda - 2 \end{vmatrix} = 0$$

$$R_2 \rightarrow R_2 - R_1$$

$$\begin{vmatrix} -1 & -3 & 4 \\ 0 & 0 & 2\lambda - 6 \\ \lambda + 1 & \lambda + 2 & -\lambda - 2 \end{vmatrix} = 0$$

$$(2\lambda - 6)(-\lambda - 2 + 3\lambda + 3) = 0$$

$$(\lambda - 3)(2\lambda + 1) = 0$$

$$\lambda = 3 \text{ or } \lambda = -\frac{1}{2}$$

$$\lambda^2 + \lambda = 9 + 3 = 12$$

49. Let $A = \{1, 6, 11, 16, \dots\}$ and $B = \{9, 16, 23, 30, \dots\}$ be the sets consisting of the first 2025 terms of two arithmetic progressions. Then $n(A \cup B)$ is :

माना $A = \{1, 6, 11, 16, \dots\}$ और $B = \{9, 16, 23, 30, \dots\}$ दो समुच्चय हैं, जिनमें दो समान्तर श्रेढ़ियों के प्रथम 2025 पद हैं। तो $n(A \cup B)$ है :

- (A) 3761 (B) 4027 (C) 4003 (D) 3814

Ans. A

Sol.
$$\left. \begin{array}{l} A = \{1, 6, 11, 16, \dots\} \\ B = \{9, 16, 23, 30, \dots\} \end{array} \right\} \rightarrow 2025 \text{ terms}$$

First common term = 16

$$d_1 = 5, d_2 = 7$$

$$\text{LCM}(d_1, d_2) = 35$$

So, common terms sequence will be $\{16, 51, 86, \dots\}$

Now, 2025th term of AP 'A'

$$1 + (2025 - 1) \cdot 5 = 10121$$

$$\therefore 16 + (n - 1) \cdot 35 \leq 10121$$

$$n - 1 \leq 288.71$$

$$n \leq 289.71$$

$$n = 289$$

50. If the system of equations $x + 2y - 3z = 2$

$$2x + \lambda y + 5z = 5$$

$$14x + 3y + \mu z = 33$$

has infinitely many solutions, then $\lambda + \mu$ is equal to :

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$$x + 2y - 3z = 2$$

$$2x + \lambda y + 5z = 5$$

$$14x + 3y + \mu z = 33$$

के अनन्त हल हैं तो $\lambda + \mu$ बराबर है :

- (A) 13 (B) 12 (C) 10 (D) 11

Ans. B

Sol.
$$D = \begin{vmatrix} 1 & 2 & -3 \\ 2 & \lambda & 5 \\ 14 & 3 & \mu \end{vmatrix} = \lambda\mu - 4\mu + 42\lambda + 107$$

$$D_x = \begin{vmatrix} 2 & 2 & -3 \\ 5 & \lambda & 5 \\ 33 & 3 & \mu \end{vmatrix} = 2\lambda\mu - 10\mu + 99\lambda + 255$$

$$D_y = \begin{vmatrix} 1 & 2 & -3 \\ 2 & 5 & 5 \\ 14 & 33 & \mu \end{vmatrix} = \mu - 13$$

$$D_z = \begin{vmatrix} 1 & 2 & 2 \\ 2 & \lambda & 5 \\ 14 & 3 & 33 \end{vmatrix} = 5\lambda + 5$$

for infinite solutions $D = D_x = D_y = D_z = 0$

$$\therefore \lambda = -1, \mu = 13$$

51. $1 + 3 + 5^2 + 7 + 9^2 + \dots$ upto 40 terms is equal to :

$1 + 3 + 5^2 + 7 + 9^2 + \dots$ का 40 पदों तक योग है :

(A) 33980 (B) 41880 (C) 43890 (D) 40870

Ans. B

Sol. $1^2 + 3 + 5^2 + 7 + 9^2 + 11 + 13^2 + 15 + \dots 77^2 + 79$ upto 40 terms

$$a_n = 1 + (40 - 1)2$$

$$= 1 + 39.2$$

$$= 79$$

$$\{1^2 + 5^2 + 9^2 + 13^2 + \dots + 77^2\} + \{3 + 7 + 11 + \dots + 79\}$$

↓

20 terms

↓

20 terms

$$\sum_{r=1}^{20} (4r-3)^2 + \sum_{r=1}^{20} (4r-1)$$

$$\sum_{r=1}^{20} (16r^2 - 24r + 9) + 4 \sum_{r=1}^{20} r - 20$$

$$16 \sum_{r=1}^{20} r^2 - 24 \sum_{r=1}^{20} r + 9.20 + 4 \sum_{r=1}^{20} r - 20$$

$$\frac{16 \times 20 \times 21 \times 41}{6} - \frac{20 \times 20 \times 21}{2} + 8 \times 20$$

$$45920 - 4200 + 160$$

$$\Rightarrow 41880$$

52. If $\log_e a, \log_e b, \log_e c$ are in A.P. and $\log_e a - \log_e 2b, \log_e 2b - \log_e 3c, \log_e 3c - \log_e a$ are also in an A.P., then $a : b : c$ is equal to:

यदि $\log_e a, \log_e b, \log_e c$ एक A.P. में है तथा $\log_e a - \log_e 2b, \log_e 2b - \log_e 3c, \log_e 3c - \log_e a$ भी एक A. P. में है, तो $a : b : c$ बराबर है :

- (A) $9 : 6 : 4$ (B) $25 : 10 : 4$ (C) $6 : 3 : 2$ (D) $16 : 4 : 1$

Ans. A

Sol. $\log_e a, \log_e b, \log_e c$ are in A.P.

$$\therefore b^2 = ac$$

Also $\log_e \left(\frac{a}{2b} \right), \log_e \left(\frac{2b}{3c} \right), \log_e \left(\frac{3c}{a} \right)$ are in A.P.

$$\left(\frac{2b}{3c} \right)^2 = \frac{a}{2b} \times \frac{3c}{a}$$

$$\frac{b}{c} = \frac{3}{2}$$

$$\text{Putting in eq. (i) } b^2 = a \times \frac{2b}{3}$$

$$\frac{a}{b} = \frac{3}{2}$$

$$a : b : c = 9 : 6 : 4$$

53. For $0 < c < b < a$, let $(a + b - 2c)x^2 + (b + c - 2a)x + (c + a - 2b) = 0$ and $\alpha \neq 1$ be one of its root. Then, among the two statements :

(I) If $\alpha \in (-1, 0)$, then b cannot be the geometric mean of a and c .

(II) If $\alpha \in (0, 1)$, then b may be the geometric mean of a and c .

- (A) Neither (I) nor (II) is true. (B) Only (I) is true
(C) Only (II) is true. (D) Both (I) and (II) are true

$0 < c < b < a$, के लिए माना $(a + b - 2c)x^2 + (b + c - 2a)x + (c + a - 2b) = 0$ का एक मूल $\alpha \neq 1$ है। तो दो दो कथनों :

(I) यदि $\alpha \in (-1, 0)$ है, तो a तथा b का गुणोत्तर माध्य b नहीं हो सकता।

(II) यदि $\alpha \in (0, 1)$ है, तो a तथा c का गुणोत्तर माध्य b हो सकता है।

(A) (I) तथा (II) दोनों सत्य हैं

(B) केवल (I) सत्य है

(C) केवल (II) सत्य है

(D) न तो (I) न ही (II) सत्य हैं

Ans. D

Sol. $(a + b - 2c)x^2 + (b + c - 2a)x + (c + a - 2b) = 0$. Hence $0 < c < b < a$

When $x = 1$

$$(a + b - 2c) + (b + c - 2a) + (c + a - 2b) = 0$$

So one root of equation is $\beta = 1$ and other root is given $\alpha \neq 1$

$$\alpha\beta = \frac{c + a - 2b}{a + b - 2c}$$

$$\therefore \alpha = \frac{c + a - 2b}{a + b - 2c}$$

when $-1 < \alpha < 0$

$$-1 < \frac{c + a - 2b}{a + b - 2c} < 0$$

$$(\because c < a, c < b, 2c < a + b)$$

$$-a - b + 2c < c + a - 2b < 0$$

$$2b > a + c$$

$$b > \frac{a + c}{2}$$

$$(G = \sqrt{ac})$$

$$(G < A)$$

So b can not be GM of a and c .

When $0 < \alpha < 1$

$$0 < \frac{c + a - 2b}{a + b - 2c} < 1$$

$$0 < c + a - 2b < a + b - 2c$$

$$2b < c + a$$

$$b < \frac{a + c}{2}; b = \sqrt{ac};$$

$$GM < AM$$

So b may be GM of a and c .

54. Let $P_n = \alpha^n + \beta^n, n \in \mathbb{N}$. If $P_{10} = 123, P_9 = 76, P_8 = 47$ and $P_1 = 1$, then the quadratic equation having roots $\frac{1}{\alpha}$

and $\frac{1}{\beta}$ is :

माना $P_n = \alpha^n + \beta^n$, $n \in \mathbb{N}$. यदि $P_{10} = 123$, $P_9 = 76$, $P_8 = 47$ और $P_1 = 1$ हैं, तो वह द्विघातीय समीकरण, जिसके मूल $\frac{1}{\alpha}$ और

$\frac{1}{\beta}$ हैं, है :

- (A) $x^2 - x - 1 = 0$ (B) $x^2 - x + 1 = 0$ (C) $x^2 + x + 1 = 0$ (D) $x^2 + x - 1 = 0$

Ans. D

Sol. $\alpha^{10} + \beta^{10} = 123$

$$\alpha + \beta = 1$$

$$\alpha^9 + \beta^9 = 76$$

$$\alpha^8 + \beta^8 = 47$$

$$P_{10} = P_9 + P_8$$

$$x^2 = x + 1 \quad \Rightarrow \quad x^2 - x - 1 = 0$$

$$\alpha + \beta = 1, \alpha\beta = -1$$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{1}{-1} = -1, \frac{1}{\alpha\beta} = -1$$

55. Let $\langle a_n \rangle$ be a sequence such that $a_0 = 0$, $a_1 = \frac{1}{2}$ and $2a_{n+2} = 5a_{n+1} - 3a_n$, $n = 0, 1, 2, 3, \dots$. Then $\sum_{k=1}^{100} a_k$ is equal to :

माना कि $\langle a_n \rangle$ एक अनुक्रम इस प्रकार है कि $a_0 = 0$, $a_1 = \frac{1}{2}$ और $2a_{n+2} = 5a_{n+1} - 3a_n$, $n = 0, 1, 2, 3, \dots$ है। तब $\sum_{k=1}^{100} a_k$

बराबर है :

- (A) $3a_{99} - 100$ (B) $3a_{99} + 100$ (C) $3a_{100} - 100$ (D) $3a_{100} + 100$

Ans. C

Sol. $a_0 = 0$, $a_1 = \frac{1}{2}$

$$2a_{n+2} = 5a_{n+1} - 3a_n$$

$$2x^2 - 5x + 3 = 0 \Rightarrow x = 1, \frac{3}{2}$$

$$\therefore a_n = A1^n + B\left(\frac{3}{2}\right)^n$$

$$\left. \begin{array}{l} n=0 \quad 0 = A + B \\ n=1 \quad \frac{1}{2} = A + \frac{3}{2}B \end{array} \right\} \begin{array}{l} A = -1 \\ B = 1 \end{array}$$

$$\Rightarrow a_n = -1 + \left(\frac{3}{2}\right)^n$$

$$\sum_{k=1}^{100} a_k = \sum_{k=1}^{100} (-1) + \left(\frac{3}{2}\right)^k$$

$$= -100 + \frac{\left(\frac{3}{2}\right)\left(\left(\frac{3}{2}\right)^{100} - 1\right)}{\frac{3}{2} - 1}$$

$$= -100 + 3\left(\left(\frac{3}{2}\right)^{100} - 1\right)$$

$$= 3 \cdot (a_{100}) - 100$$

56. Let the system of linear equations

$$x + y + kz = 2$$

$$2x + 3y - z = 1$$

$$3x + 4y + 2z = k$$

have infinitely many solutions. Then the system

$$(k+1)x + (2k-1)y = 7$$

$$(2k+1)x + (k+5)y = 10$$

has :

(A) no solution

(B) unique slution satisfying $x - y = 1$

(C) infinitely many solutions

(D) unique solution satisfying $x + y = 1$

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$$x + y + kz = 2$$

$$2x + 3y - z = 1$$

$$3x + 4y + 2z = k$$

के अनंत हल है। तो निकाय

$$(k + 1)x + (2k - 1)y = 7$$

$$(2k + 1)x + (k + 5)y = 10$$

(A) का कोई हल नहीं है

(C) के अनंत हल है

(B) केवल एक हल है जो $x - y = 1$ को संतुष्ट करता है

(D) केवल एक हल है जो $x + y = 1$ को संतुष्ट करता है

Ans. D

Sol.
$$\begin{vmatrix} 1 & 1 & k \\ 2 & 3 & -1 \\ 3 & 4 & 2 \end{vmatrix} = 0$$

$$\Rightarrow 1(10) - 1(7) + k(-1) = 0$$

$$\Rightarrow k = 3$$

For $k = 3$, 2nd system is

$$4x + 5y = 7 \quad \dots(1)$$

and $7x + 8y = 10 \quad \dots(2)$

Clearly, they have a unique solution

$$(B) - (1) \Rightarrow 3x + 3y = 3$$

$$\Rightarrow x + y = 1$$

57. Let A be a 3×3 matrix such that $|\text{adj}(\text{adj}(\text{adj} A))| = 81$. If $S = \left\{ n \in \mathbb{Z} : (|\text{adj}(\text{adj} A)|)^{\frac{(n-1)^2}{2}} = |A|^{(3n^2-5n-4)} \right\}$, then

$$\sum_{n \in S} |A|^{(n^2+n)} \text{ is equal to :}$$

माना एक 3×3 आव्यूह A के लिए $|\text{adj}(\text{adj}(\text{adj} A))| = 81$ है। यदि $S = \left\{ n \in \mathbb{Z} : (|\text{adj}(\text{adj} A)|)^{\frac{(n-1)^2}{2}} = |A|^{(3n^2-5n-4)} \right\}$ है, तो

$$\sum_{n \in S} |A|^{(n^2+n)} \text{ बराबर है :}$$

(A) 820

(B) 750

(C) 866

(D) 732

Ans. D

Sol. $|\text{adj}(\text{adj}(\text{adj} A))| = 81$

$$\Rightarrow |\text{adj} A|^4 = 81$$

$$\Rightarrow |\text{adj} A| = 3$$

$$\Rightarrow |A|^2 = 3$$

$$\Rightarrow |A| = \sqrt{3}$$

$$\left(|A|^4\right)^{\frac{(n-1)^2}{2}} = |A|^{3n^2-5n-4}$$

$$\Rightarrow 2(n-1)^2 = 3n^2 - 5n - 4$$

$$\Rightarrow 2n^2 - 4n + 2 = 3n^2 - 5n - 4$$

$$\Rightarrow n^2 - n - 6 = 0$$

$$\Rightarrow (n-3)(n+2) = 0$$

$$\Rightarrow n = 3, -2$$

$$\sum_{n \in S} |A^{n^2+n}|$$

$$= |A^2| + |A^{12}|$$

$$= 3 + 36 = 3 + 729 = 732$$

58. If the sum of the second, fourth and sixth terms of a G.P. of positive terms is 21 and the sum of its eighth, tenth and twelfth terms is 15309, then the sum of its first nine terms is :

यदि धनात्मक पदों की एक G.P. के दूसरे, चौथे और छठे पदों का योग 21 है तथा इसके आठवें, दसवें और बारहवें पदों का योग 15309 है, तो इसके प्रथम नौ पदों का योग है :

(A) 755

(B) 757

(C) 760

(D) 750

Ans. B

Sol. $ar + ar^3 + ar^5 = 21 \quad \dots(1)$

$ar^7 + ar^9 + ar^{11} = 15309 \quad \dots(2)$

$$(B) \div (1) \quad \frac{ar^7(1+r^2+r^4)}{ar(1+r^2+r^4)} = \frac{15309}{21} = 729$$

$$\Rightarrow r^6 = 729 \Rightarrow r = 3 \text{ (terms are positive)}$$

$$\text{and from (1) } a = \frac{1}{13}$$

Sum of first nine terms

$$= a(1 + r + r^2 + \dots + r^8)$$

$$= \frac{1}{13} \frac{(3^9 - 1)}{2}$$

$$= 757$$

59. For the system of linear equations

$$x + y + z = 6$$

$$\alpha x + \beta y + 7z = 3$$

$$x + 2y + 3z = 14,$$

which of the following is **NOT** true?

(A) If $\alpha = \beta = 7$, then the system has no solution.

(B) If $\alpha = \beta$ and $\alpha \neq 7$, then the system has a unique solution.

(C) There is a unique point (α, β) on the line $x + 2y + 18 = 0$ for which the system has infinitely many solutions.

(D) For every point $(\alpha, \beta) \neq (7, 7)$ on the line $x - 2y + 7 = 0$, the system has infinitely many solutions.

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$$x + y + z = 6$$

$$\alpha x + \beta y + 7z = 3$$

$$x + 2y + 3z = 14,$$

के लिए निम्न में से कौन सा सत्य नहीं है ?

(A) यदि $\alpha = \beta = 7$ है, तो निकाय का कोई हल नहीं है

(B) यदि $\alpha = \beta$ तथा $\alpha \neq 7$ है, तो निकाय का केवल एक हल है

(C) रेखा $x + 2y + 18 = 0$ पर केवल एक बिंदु (α, β) है जिसके लिए निकाय के अनंत हल हैं

(D) रेखा $x - 2y + 7 = 0$ पर प्रत्येक बिंदु $(\alpha, \beta) \neq (7, 7)$ के लिए निकाय के अनंत हल हैं

Ans. D

Sol. By equation 1 and 3 $y + 2z = 8$

$$y = 8 - 2z$$

$$\text{And } x = -2 + z$$

Now putting in equation 2

$$\alpha(z - 2) + \beta(-2z + 8) + 7z = 3$$

$$\Rightarrow (\alpha - 2\beta + 7)z = 2\alpha - 8\beta + 3$$

So equations have unique solution if

$$\alpha - 2\beta + 7 \neq 0$$

And equations have no solution if

$$\alpha - 2\beta + 7 = 0 \text{ and } 2\alpha - 8\beta + 3 \neq 0$$

And equations have infinite solution if

$$\alpha - 2\beta + 7 = 0 \text{ and } 2\alpha - 8\beta + 3 = 0$$

60. For the system of linear equations

$$2x + 4y + 2az = b$$

$$x + 2y + 3z = 4$$

$$2x - 5y + 2z = 8$$

which of the following is NOT correct?

(A) has unique solution if $a = b = 8$

(B) It has infinitely many solutions if $a = 3, b = 8$

(C) It has infinitely many solutions if $a = 3, b = 6$

(D) It has unique solution if $a = b = 6$

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$$2x + 4y + 2az = b$$

$$x + 2y + 3z = 4$$

$$2x - 5y + 2z = 8$$

के लिए निम्न में से कौनसा सही नहीं है?

(A) इसका अद्वितीय हल है यदि $a = b = 8$ हैं

(B) इसके अनंत हल हैं यदि $a = 3, b = 8$ हैं

(C) इसके अनंत हल हैं यदि $a = 3, b = 6$ हैं

(D) इसका अद्वितीय हल है यदि $a = b = 6$ हैं

Ans. C

Sol.
$$D = \begin{vmatrix} 2 & 4 & 2a \\ 1 & 2 & 3 \\ 2 & -5 & 2 \end{vmatrix}$$

$$= 2(4 + 15) - 4(2 - 6) + 2a(-5 - 4)$$

$$= 38 + 16 - 18a = 54 - 18a = 18(3 - a)$$

$$D_1 = \begin{vmatrix} b & 4 & 2a \\ 4 & 2 & 3 \\ 8 & -5 & 2 \end{vmatrix}$$

$$= b(4 + 15) - 4(8 - 24) + 2a(-20 - 16)$$

$$= 19b + 64 - 72a$$

$$= 19b - 72a + 64$$

$$D_2 = \begin{vmatrix} 2 & b & 2a \\ 1 & 4 & 3 \\ 2 & 8 & 2 \end{vmatrix}$$

$$= 2(8 - 24) - b(2 - 6) + 2a(8 - 8)$$

$$= -32 + 4b = 4(b - 8)$$

$$D_3 = \begin{vmatrix} 2 & 4 & b \\ 1 & 2 & 4 \\ 2 & -5 & 8 \end{vmatrix}$$

$$= 2(16 + 20) - 4(8 - 8) + b(-5 - 4)$$

$$= 72 - 9b = 9(8 - b)$$

Now when $a \neq 3$, $b \in \mathbb{R}$ system have unique solution.

when $a = 3$, $b \neq 8$, system have no solution

when $a = 3$, $b = 8$, system have infinitely many solutions.